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Zan Ahmad

Department of Applied Mathematics & Statistics
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Education

Johns Hopkins University, Whiting School of Engineering — June 2022 – May 2027 (expected)
Ph.D. Candidate, Applied Mathematics & Statistics
Advisors: Natalia Trayanova, Mauro Maggioni

Johns Hopkins University, Whiting School of Engineering — June 2022 – May 2025
M.S.E., Applied Mathematics & Statistics

New York University, Courant Institute of Mathematical Sciences — Sept 2017 – Dec 2020
B.A., Mathematics
Minors: Chemistry, Film Production
Advisors: Charles S. Peskin, Charles Puelz

Research Interests

Geometric machine learning · Riemannian shape analysis · Neural operators for PDEs · Cardiovascular modeling

Research Experience

Johns Hopkins University, Institute of Computational Medicine — June 2022 – Present
Graduate Researcher, Computational Cardiology Lab
Advisors: Natalia Trayanova, Mauro Maggioni

Yale School of Medicine, Department of Neurology — June 2021 – May 2022
Postgraduate Researcher, Blumenfeld Lab
Advisor: Hal Blumenfeld

New York University, Center for Neural Science — Sept 2020 – May 2021
NIH Training Program in Computational Neuroscience
Advisor: Charles S. Peskin

New York University, Courant Institute of Mathematical Sciences — June 2020 – Aug 2020
Applied Mathematics Summer Undergraduate Research Experience (AM-SURE)
Advisors: Charles Puelz, Charles S. Peskin

New York University, Courant Institute of Mathematical Sciences — Sept 2019 – May 2020
Modeling & Simulation Research Training Group
Advisor: Charles Puelz

New York University, Department of Chemistry — Jan 2018 – Aug 2019
Undergraduate Researcher, DNA Nanotechnology Lab
Advisors: Nadrian Seeman, Yoel P. Ohayon

Publications & Preprints

[†] *equal contribution.* Full list on Google Scholar.

Published & Accepted

1. "A Computational Pipeline for Clustering Left Atrial Appendage Morphology via Elastic Shape Analysis."
   Z. Ahmad, M. Yin, Y. Sukurdeep, N. Rotenberg, E. Kholmovski, N. Trayanova.
   Computers in Biology and Medicine, 196:110905, 2025.
2. "A Comprehensive Stroke Risk Assessment by Combining Atrial Computational Fluid Dynamics Simulations and Functional Patient Data."
   A. Zingaro, **Z. Ahmad**, E. Kholmovski, K. Sakata, L. Dede', A.K. Morris, A. Quarteroni, N. Trayanova.
   Scientific Reports, 14:9515, 2024.
3. "Optimal Fenestration of the Fontan Circulation."
   Z. Ahmad, L.H. Jin, D.J. Penny, C. Rusin, C.S. Peskin, C. Puelz.
   Frontiers in Physiology: Computational Physiology and Medicine, 13, 2022.
4. "Thinner Latent Spaces: Detecting Dimension and Imposing Invariance with Conformal Autoencoders."
   G.A. Kevrekidis, **Z. Ahmad**, M. Maggioni, S. Villar, Y.G. Kevrekidis.
   SIAM Journal on Mathematics of Data Science, accepted.
5. "Steady-State Analysis of Gravitational Effects on Hemodynamics."
   A. Kennard[†], **Z. Ahmad**[†], J. Solak, J.W. Davis, R.S. Blue, C.S. Peskin, K.M. Ong.
   Scientific Reports, accepted.

Workshop Papers

1. "Diffeomorphic Latent Neural Operators for Data-Efficient Learning of Solutions to Partial Differential Equations."
   Z. Ahmad, S. Chen, M. Yin, A. Kumar, N. Charon, N. Trayanova, M. Maggioni.
   4th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE), 2025.
2. "Convolutional Deep Operator Networks for Learning Nonlinear Focused Ultrasound Wave Propagation in Heterogeneous Spinal Cord Anatomy."
   A. Kumar, X. Zhi, **Z. Ahmad**, M. Yin, A. Manbachi.
   4th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE), 2025.

Under Review & Preprints

1. "Graph Fourier Neural Kernels: Learning Full PDE Trajectories Across Geometries and Anisotropic Parameter Fields."
   S. Loeffler, **Z. Ahmad**, S. Chen, S.Y. Ali, C. Yamamoto, K. Sakata, D.M. Popescu, A. Yee, Y. Lal, N. Trayanova, M. Maggioni.
   arXiv preprint, 2026. *Under review, PNAS.*
2. "Predicting the *When*: Multimodal AI for Time-to-Recurrence Analysis After Atrial Fibrillation Ablation."
   M. Yin, C. Lai, R. Yadav, . . . , **Z. Ahmad**, S. Chen, . . . , N. Trayanova.
   medRxiv preprint, 2026.
3. "Stimulation of the Thalamus for Arousal Restoral in Temporal Lobe Epilepsy (START) Clinical Trial: Modulation of Natural Sleep and Focal Seizures."
   T. Yadav, Z. Zhang, V. Kremen, **Z. Ahmad**, . . . , H. Blumenfeld.
   medRxiv preprint, 2025.
4. "Low and High Frequency Signatures of Impaired Consciousness in Temporal Lobe Seizures."
   T. Yadav, . . . , **Z. Ahmad**, . . . , H. Blumenfeld.
   bioRxiv preprint, 2025.

Publications & Preprints (continued)

In Preparation

1. "Varifold-Based Cardiac Shape Analysis for Generalizable and Scalable Encoding of Arrhythmia-Related Phenotypes."
   Z. Ahmad†, E. Hartman†, M. Yin†, Y. Sukurdeep, C. Yamamoto Alves Pinto, S. Chen, J. Noh, K. Sakata, A. Prakosa, E. Kholmovski, J. Chrispin, D. Spragg, H. Calkins, K. Aronis, N. Charon, M. Maggioni, N. Trayanova.
   Manuscript in preparation.
2. "Toward a mechanistic understanding of control during simulated blood loss using compartmental models and time-frequency analysis."
   C. Puelz, B.E. Griffith, J.L. Marzuola, **Z. Ahmad**, C.S. Peskin, H.-T. Wu.
   Manuscript in preparation.
3. "The Lattice Boltzmann Method for Scalable Cardiac Electrophysiology: From Theory to Patient-Specific Digital Twins."
   R. Brody†, S. Loeffler†, K. Zhang, J. Tso, C. Yamamoto Alves Pinto, **Z. Ahmad**, K. Sakata, S.Y. Ali, A. Prakosa, Y. Moshen, J. Chrispin, E. Kholmovski, M. Maggioni, N. Trayanova.
   Manuscript in preparation.

Teaching

Teaching portfolio and materials

Introduction to Scientific Programming in Python	W 2027	JHU	Instructor of Record
Probability & Statistics for Life Sciences *Four-credit departmental course; lectures, office hours, and exam grading within a shared curriculum* Materials ↗	F 2026	JHU	Instructor of Record
Introduction to Scientific Programming in Python *Scientific libraries, numerical linear algebra, optimization, ML (neural networks, CNNs, transformers, LLMs); algorithmic-thinking and NumPy assignment* Assignment ↗ · Materials ↗	W 2026	JHU	Instructor of Record
Biomedical Engineering Innovation & Design	Su 2025	JHU	Instructor of Record
Biomedical Engineering Innovation & Design	Sp 2025	JHU	Instructor of Record
Introduction to Scientific Programming in Python Materials ↗	W 2025	JHU	Instructor of Record
Advanced Topics in Computational Cardiology *0D/1D modeling, shape analysis, manifold learning, computer vision, neural operators; upper-level seminar. Heart-rate feedback-control project; finite-difference Python tutorial* Feedback control ↗ · Python tutorial ↗ · Materials ↗	F 2024	JHU	Creator; Instructor of Record
Mathematical Modeling & Computer Simulation *First-principles biological modeling; course projects served as launching points for undergraduate research mentorship* Materials ↗	F 2024	JHU	Creator; Instructor of Record
Biomedical Engineering Innovation & Design	F 2024	JHU	Instructor of Record
Biomedical Engineering Innovation & Design	Su 2024	JHU	Instructor of Record
Mathematical Cardiology *First-principles cardiovascular modeling as an accessible on-ramp to research in mathematical biology and cardiovascular modeling* Materials ↗	F 2023	JHU	Creator; Instructor of Record
Biomedical Engineering Innovation & Design	Su 2023	JHU	Instructor of Record

Teaching (continued)

Computational Neuroscience: Channels & Networks *Developed MATLAB tutorials on neuronal action potential models; students used these as a foundation for final course projects. [tutorial videos]* Materials ↗	Sp 2021	NYU	Guest Lecturer
Linear Algebra	F 2020	NYU	Grader
Calculus I	Sp 2020	NYU	Grader
Calculus I	F 2019	NYU	Grader
Thermodynamics; Organic Chemistry I & II *F 2019: Thermodynamics and Organic Chemistry II; Sp 2019: Organic Chemistry I*	Sp–F 2019	NYU	Lead Chemistry Tutor
Physics *East Harlem Tutorial Program: nonprofit providing free tutoring and mentorship to K–12 students in East Harlem, NY*	2018–19	EHTP	Volunteer Tutor

Mentoring

Shiyi Chen · M.S. student, Mechanical Engineering, Johns Hopkins University — Apr 2024 – Aug 2025
Thesis: *Learning PDE solution operators via diffeomorphic mappings: applications in fluid dynamics.*
Next position: ***Yale University****, Biomedical Engineering, Ph.D. student.*

Noam Rottenberg · Undergraduate, Engineering, Johns Hopkins University — Aug 2023 – May 2024
Topics: Automatic left atrial segmentation and preprocessing for statistical shape analysis.
Next position: ***Yale School of Medicine****, Medical student.*

Ayush Mandal · Undergraduate, Applied Mathematics & Computational Neuroscience, JHU — Sept 2023 – Dec 2024
Topics: Feedback control of heart rate in zero-dimensional circulation models; modeling neuronal dynamics.

Industry Experience

Intuit Inc., AI/Data Science Department · San Diego, CA — June 2025 – Sept 2025
AI Science Intern, Consumer Group Analytics and Decision Science
Advisor: Dr. Joseph Powers
- Implemented a Bayesian guardrail-testing framework in a decision-making tool used across Intuit departments by approximately 5,000 users; analysis of TY23–TY24 data informed TY25 decisions with an estimated $1.2 million financial impact.

Virgin Galactic · Remote — Jan 2023 – Aug 2023
Computational Research Intern
Advisor: Dr. Karen Ong
- Built and calibrated circulatory models of hypergravity using G-force data from centrifuge simulations.

Grants, Awards & Honors

- **American Heart Association Predoctoral Fellowship** · $67,388 — July 2024 – June 2026
  Quantifying Stroke Risk in Atrial Fibrillation with Computational Fluid Dynamics and Machine Learning.
- **NIH Cardiovascular Epidemiology Training Grant** (T32HL007024) · $27,400 — July 2023 – June 2024
  Fellowship for PhD students with research focused on cardiovascular disease.
- **NIH Computational Medicine Training Grant** (T32GM119998) · $25,836 — June 2022 – May 2023
  Fellowship for PhD students at the Institute of Computational Medicine, JHU.

- **NIH Computational Neuroscience Training Grant** (R90DA043849) · $13,336 — Aug 2020 – May 2021
  Awarded to 6 of 50 applicants. *Short time-scale dynamics in a computational model of signal propagation in a cerebellar neuronal network.*
- **NSF Modeling & Simulation RTG** (DMS-1646339) · $6,250 — May – Aug 2020
  AM-SURE at NYU Courant Institute. Three undergraduates selected from 60 applicants.
- **Acheson J. Duncan Fund Travel Award** · $3,150 — April 2023
  Fund for the Advancement of Research in Statistics. Supported attendance at the Cardiac Physiome Conference, Irvine, CA, 2023.
- **Wilfred L. & Ruth S.F. Peltz Research Scholar Award** · $2,750 — May 2020
  Awarded for NYU DURF proposal: *Optimally fenestrated Fontan circulation.*
- **IEEE EMBS Student Fellowship** · $2,000 — June 2024
  One of 14 selected from 900 applicants. Full scholarship for AI in Medicine & Healthcare workshop. Chania, Crete, Greece.
- **Heart Rhythm Society (HRS) Travel Grant** · $1,000
  Supported presentation of the stroke risk project.
- **Dean's Undergraduate Research Fund Grant (DURF)** · $1,000 — May 2020
  Optimal fenestration of Fontan circulation. NYU College of Arts and Sciences.
- **Dean's Undergraduate Research Fund Training Grant** · $500 — May 2018
  Self-assembly of A-form 3D DNA crystals of 2'-fluoro DNA.
- **IDIES Poster Competition — 1st Place** · $100 — Oct 2023
  Random Convolutional Features and Patch-Based Learning for Multitask Image Classification. [poster]
- **Conference Travel Support, NYU Courant Modeling & Simulation Research Training Group (RTG)** — 2022–2023
  Supported research presentations at the SIAM Annual and Life Sciences Conferences, Pittsburgh (July 2022), and the ECCOMAS Young Investigators Conference, Porto, Portugal (June 2023).

Conference Presentations

- **Lead organizer**, inaugural SIAM DMV Area Applied Math Conference, Johns Hopkins University, Baltimore, MD, April 26, 2025.
  Event highlights (Instagram) · Conference photo album · Conference flyer

1. "AI-Based Shape Analysis Correlates Left Atrial Appendage Morphology with Stroke Risk in Patients with Atrial Fibrillation."
   Heart Rhythm Society (HRS), San Diego, CA, April 2025.
2. "Multimodal Artificial Intelligence for Predicting Atrial Fibrillation Recurrence Using Pre-Procedural Data."
   Heart Rhythm Society (HRS), San Diego, CA, April 2025.
3. "Mathematical Cardiology: Operator Learning, Shape Analysis, and Computational Fluid Dynamics."
   IEEE EMBS Leadership Academy Workshop on AI/Data Science in Medicine and Healthcare, Chania, Crete, Greece, June 2024.
4. "Random Convolutional Features and Patch-Based Learning for Multitask Image Classification."
   IDIES Annual Research Symposium, Baltimore, MD, October 2023. **1st Place Best Poster Award.** [poster]
5. "Random Convolutional Features and Patch-Based Learning for Multitask Image Classification."
   Johns Hopkins University AI-X Symposium, Baltimore, MD, September 2023. [poster]
6. "Non-Pulsatile ODE Model of Circulation Response to Hypergravity."
   SIAM Conference on Control Theory, Philadelphia, PA, July 2023.
7. "Modeling Feedback Control Response to Hemodynamic Perturbations in a Pulsatile Circulation Model."
   SIAM Conference on Control Theory, Philadelphia, PA, July 2023.
8. "Hemodynamics and Oxygen Model for the Fontan Circulation with Feedback Control in Variable Heart Rate."
   ECCOMAS Minisymposium on Multiphysics and Multiscale Simulations in Cardiovascular Applications, Porto, Portugal, June 2023.
9. "A Dynamic Mathematical Model for the Prediction of Cardiovascular Response to Hypergravity."
   American Society of Aerospace Medicine (ASMA), New Orleans, LA, May 2023.
10. "Stroke Risk Is Identified by Slow Blood Flow in the Left Atrium."
    Heart Rhythm Society (HRS), New Orleans, LA, May 2023.
11. "Modeling the Interplay Between Blood Flow and Oxygen Transport in the Defective Heart."
    SIAM Conference on Dynamical Systems, New Orleans, LA, May 2023.

12. "Hemodynamic Indicators of Stroke Risk Uncovered by Personalized Fluid Dynamics Models." *Cardiac Physiome Workshop*, Irvine, CA, May 2023.
13. "A Dynamic Model for the Control of Cardiovascular Response to Spaceflight." *International Conference of Aerospace Medicine*, Paris, France, September 2022.
14. "Simulation of Circulation Response to Gravitational Changes During Spaceflight." *SIAM Conference on Life Sciences*, Pittsburgh, PA, July 2022.
15. "Optimal Fenestration of the Fontan Circulation." *SIAM Annual Meeting*, Pittsburgh, PA, July 2022. [report]
16. "Mathematical Modeling of Cardiac Circulation Under Hypergravity." *American Society of Aerospace Medicine (ASMA)*, Reno, NV, May 2022.
17. "Increased Intracranial EEG Power and Duration in Temporal Lobe Seizures with Impaired Consciousness." *American Epilepsy Society*, Chicago, IL, December 2021.
18. "Mechanisms of Impaired Consciousness in Medial Temporal Lobe Seizures Investigated with Intracranial EEG." *Society for Neuroscience (SfN)*, Chicago, IL, November 2021.
19. "Mathematical Model for an Optimally Fenestrated Fontan Circulation." *Undergraduate Mathematics Symposium, University of Illinois at Chicago*, Chicago, IL, December 2021.

Seminars & Invited Talks

1. "A Scalable Approach to Riemannian Geometric Statistics for Cardiovascular Shape Modeling." *Mostly Biomathematics Lunchtime Seminar*, NYU Courant Institute of Mathematical Sciences, April 2026. [abstract]
2. "Neural Operators for Learning PDEs on Geometrically Varying Data." *JHU Applied Mathematics & Statistics Department Seminar*, April 2024 & December 2024.
3. "Hemodynamic Indicators of Stroke Risk Uncovered via Personalized Fluid Dynamics Simulations." *JHU Applied Mathematics & Statistics Department Seminar*, September 2023.
4. "Steady-State Analysis of Gravitational Effects on Hemodynamics." *JHU Applied Mathematics & Statistics Department Seminar*, February 2023.
5. "Pulsatile Model for the Fontan Circulation and a Feedback Control Method for Heart Rate." *JHU Applied Mathematics & Statistics Department Seminar*, November 2022.
6. "Short Time-Scale Dynamics in a Computational Model of Signal Propagation in a Cerebellar Neural Network." *NYU Training Program for Computational Neuroscience Conference*, June 2021.
7. "Using Mathematics to Study and Optimize Congenital Heart Disorders." *NYU College of Arts and Sciences Undergraduate Research Conference*, May 2021.
8. "Feedback Control of Blood Pressure and Heart Rate in a Pulsatile Cardiovascular Model." *NYU Courant Institute Modeling and Simulation Research Training Group Seminar*, March 2021.
9. "Mathematical Model for an Optimally Fenestrated Fontan Circulation." *NYU Courant Institute Applied Mathematics Research Presentations*, October 2020.

Service & Leadership

- Founder and President, **JHU SIAM Student Chapter**. Lead organizer of inaugural SIAM DMV Area Applied Math Conference, April 2025 — full-day event with UMDCP, UMBC, GWU, GU, GMU, and JHU, over 100 attendees.
  Event highlights (Instagram) · Conference photo album · Conference flyer
- Editor-in-Chief, **NYU Medical Dialogue Review**.
- Volunteer Physics Tutor, **East Harlem Tutorial Program (EHTP)** — January 2019 Volunteer of the Month. EHTP provides free after-school academic support, mentoring, and college preparation for students in East Harlem.

Media Publications

1. "Circulation Models Assess the Impacts of Congenital Heart Defects and Hypergravity." Jillian Kunze, *SIAM News*, August 2023. *Coverage of SIAM Control Theory Conference presentation.*
2. "Medicine and Biology Through the Lens of Mathematics." **Z. Ahmad**, *NYU Medical Dialogue Review*, Vol. XIV, Issue 2, pp. 44–47, April 2020.